

JESS Thermodynamic Database v8.9

Chemical species in reactions with Tantalum

24-Jan-23

Charge	CAS	Count	Molecular formula	Mol. mass
Ascorbic-2 L-Ascorbate ion; Ascorbate ion				
-2	63983-50-6	133	C(6)H(6)O(6)	174.110
B(s) Boron; Boron, rhombohedral; Boron, beta				
0	7440-42-8	111	B(1)	10.8100
Br2(l) Bromine, liquid				
0	---	285	Br(2)	159.808
C(s) Carbon; Graphite; Carbon, hexagonal				
0	7440-44-0	530	C(1)	12.0110
Cl2(g) Chlorine gas				
0	7782-50-5	589	Cl(2)	70.9060
Cs(s) Caesium; Cesium; Caesium, cubic				
0	7440-46-2	37	Cs(1)	132.910
e-1 Electron				
-1	---	3399	E(1)	0.000000
EDTA-4 Ethylenediaminetetraacetate anion				
-4	---	1457	C(10)H(12)N(2)O(8)	288.214
F-1 Fluoride ion				

-1	16984-48-8	1090	F (1)	18.9984
F2 (g) Fluorine gas				
0	7782-41-4	259	F (2)	37.9968
H+1 Hydrogen ion; Proton				
1	12408-02-5	31679	H (1)	1.00794
H+1_F-1				
0	---	142	F (1) H (1)	20.0063
H+1_Ta6O19-8				
-7	---	5	H (1) O (19) Ta (6)	1390.70
H+1 (-1)_Tartaric-2				
-3	---	6	C (4) H (3) O (6)	147.064
H+1 (-2)_Tartaric-2				
-4	---	3	C (4) H (2) O (6)	146.056
H+1 (2)_Ta6O19-8				
-6	---	5	H (2) O (19) Ta (6)	1391.70
H+1 (3)_Ta6O19-8				
-5	---	1	H (3) O (19) Ta (6)	1392.71
H+1 (4)_EDTA-4 Edetic Acid; EDTA; Ethylenediaminetetraacetic acid; N,N'-1,2-Ethanediyldis[N-(carboxymethyl)glycine; Ethylenedinitrilotetraacetic acid; 1,2-Diamino-N,N,N',N'-ethane tetraacetic acid; Versene Acid; (1,2-Ethanediyldinitrilo)tetraacetic acid; H4edta; Sequestic acid				
0	60-00-4	13	C (10) H (16) N (2) O (8)	292.246
H2O Water				
0	7732-18-5	5947	H (2) O (1)	18.0153

I2 (s) Iodine; Iodine, orthorhombic				
0	7553-56-2	244	I (2)	253.800
K (s) Potassium; Potassium, cubic				
0	7440-09-7	121	K (1)	39.0980
N2 (g) Nitrogen gas; Nitrogen				
0	7727-37-9	566	N (2)	28.0134
Na (s) Sodium; Sodium, cubic				
0	7440-23-5	209	Na (1)	22.9900
O2 (g) Oxygen gas; Oxygen				
0	7782-44-7	2559	O (2)	31.9988
OH-1 Hydroxide ion				
-1	14280-30-9	6558	H (1) O (1)	17.0073
Oxalic-2 Oxalate ion				
-2	338-70-5	1241	C (2) O (4)	88.0196
Rb (s) Rubidium; Rubidium, cubic				
0	7440-17-7	29	Rb (1)	85.4680
S (s) Sulphur, alpha; Sulphur, orthorhombic; Sulphur; Sulfur, alpha; Sulfur, orthorhombic; Sulfur; Sulfur-Rhmb				
0	7704-34-9	619	S (1)	32.0600
Ta+2				
2	---	2	Ta (1)	180.950

Ta+3 Tantalum(III) ion				
3	---	4	Ta (1)	180.950
Ta+3_Cl-1(3)_ (s) Tantalum trichloride				
0	---	1	Cl (3) Ta (1)	287.309
Ta+4_Cl-1(4)_ (s) Tantalum(IV) chloride; Tantalum tetrachloride				
0	---	1	Cl (4) Ta (1)	322.762
Ta+4_S-2(2)_ (s) Tantalum disulfide				
0	---	1	S (2) Ta (1)	245.070
Ta+5 Tantalum(V) ion				
5	---	55	Ta (1)	180.950
Ta+5_Ascorbic-2				
3	---	1	C (6) H (6) O (6) Ta (1)	355.060
Ta+5_Br-1(5)_ (s) Tantalum(V) bromide; Tantalum pentabromide				
0	---	1	Br (5) Ta (1)	580.470
Ta+5_Cl-1(3)_ (s)				
2	---	1	Cl (3) Ta (1)	287.309
Ta+5_Cl-1(4)_ (s)				
1	---	1	Cl (4) Ta (1)	322.762
Ta+5_Cl-1(5)_ (s) Tantalum(V) chloride; Tantalum pentachloride				
0	---	1	Cl (5) Ta (1)	358.215
Ta+5_Cs+1_Cl-1(6)_ (s)				
0	---	1	Cl (6) Cs (1) Ta (1)	526.578

Ta+5_F-1				
4	---	1	F(1)Ta(1)	199.948
Ta+5_F-1(2)				
3	---	1	F(2)Ta(1)	218.947
Ta+5_F-1(3)				
2	---	5	F(3)Ta(1)	237.945
Ta+5_F-1(3)_OH-1(3)				
-1	---	1	F(3)H(3)O(3)Ta(1)	288.967
Ta+5_F-1(4)				
1	---	3	F(4)Ta(1)	256.944
Ta+5_F-1(5)				
0	---	5	F(5)Ta(1)	275.942
Ta+5_F-1(5)_(s) Tantalum(V) fluoride; Tantalum pentafluoride				
0	---	1	F(5)Ta(1)	275.942
Ta+5_F-1(6)				
-1	---	3	F(6)Ta(1)	294.940
Ta+5_F-1(7)				
-2	---	5	F(7)Ta(1)	313.939
Ta+5_F-1(8)				
-3	---	2	F(8)Ta(1)	332.937
Ta+5_F-1(9)				
-4	---	1	F(9)Ta(1)	351.936
Ta+5_H+1(-1)_OH-1(4)_Tartaric-2				
-2	---	1	C(4)H(7)O(10)Ta(1)	396.044

Ta+5_H+1(-2)_OH-1(4)_Tartaric-2				
-3	---	1	C(4)H(6)O(10)Ta(1)	395.036
Ta+5_H2O2				
5	---	4	H(2)O(2)Ta(1)	214.965
Ta+5_H2O2_F-1				
4	---	1	F(1)H(2)O(2)Ta(1)	233.963
Ta+5_H2O2_F-1(2)				
3	---	1	F(2)H(2)O(2)Ta(1)	252.962
Ta+5_H2O2_F-1(3)				
2	---	1	F(3)H(2)O(2)Ta(1)	271.960
Ta+5_H2O2_F-1(4)				
1	---	1	F(4)H(2)O(2)Ta(1)	290.958
Ta+5_I-1(5)_(s) Tantalum(V) iodide; Tantalum pentaiodide				
0	---	1	I(5)Ta(1)	815.450
Ta+5_K+1_Cl-1(6)_(s)				
0	---	1	Cl(6)K(1)Ta(1)	432.766
Ta+5_Na+1_Cl-1(6)_(s)				
0	---	1	Cl(6)Na(1)Ta(1)	416.658
Ta+5_OH-1				
4	---	2	H(1)O(1)Ta(1)	197.957
Ta+5_OH-1(2)				
3	---	5	H(2)O(2)Ta(1)	214.965
Ta+5_OH-1(2)_EDTA-4				
-1	---	2	C(10)H(14)N(2)O(10)Ta(1)	503.179

Ta+5_OH-1(2)_Oxalic-2				
1	---	3	C(2)H(2)O(6)Ta(1)	302.984
Ta+5_OH-1(2)_Oxalic-2(2)				
-1	---	3	C(4)H(2)O(10)Ta(1)	391.004
Ta+5_OH-1(2)_Oxalic-2(3)				
-3	---	1	C(6)H(2)O(14)Ta(1)	479.024
Ta+5_OH-1(3)				
2	---	2	H(3)O(3)Ta(1)	231.972
Ta+5_OH-1(3)_Oxalic-2_(s)				
0	---	1	C(2)H(3)O(7)Ta(1)	319.992
Ta+5_OH-1(4)				
1	---	6	H(4)O(4)Ta(1)	248.979
Ta+5_OH-1(4)_Tartaric-2				
-1	---	1	C(4)H(8)O(10)Ta(1)	397.052
Ta+5_OH-1(5)				
0	---	9	H(5)O(5)Ta(1)	265.987
Ta+5_OH-1(5)_(s)				
0	---	1	H(5)O(5)Ta(1)	265.987
Ta+5_OH-1(6)				
-1	---	1	H(6)O(6)Ta(1)	282.994
Ta+5_Rb+1_Cl-1(6)_(s)				
0	---	1	Cl(6)Rb(1)Ta(1)	479.136
Ta(g) Tantalum gas				
0	---	1	Ta(1)	180.950

Ta (s) Tantalum				
0	---	31	Ta (1)	180.950
Ta₂C (s) Ditantalum carbide				
0	---	1	C (1) Ta (2)	373.911
Ta₂N (s) Ditantalum nitride				
0	---	1	N (1) Ta (2)	375.907
Ta₂O₃ (s)				
0	---	1	O (3) Ta (2)	409.898
Ta₂O₅ (s) Tantalum(V) oxide; Ditantalum pentoxide				
0	---	5	O (5) Ta (2)	441.897
Ta₆O₁₉₋₈ polyoxotantalate				
-8	---	9	O (19) Ta (6)	1389.69
TaB₂ (s) Tantalum diboride				
0	---	1	B (2) Ta (1)	202.570
TaC (s) Tantalum monocarbide				
0	---	1	C (1) Ta (1)	192.961
TaCl_{2.5} (s) Tantalum 2.5-chloride				
0	---	1	Cl (2.5) Ta (1)	269.583
TaN (s) Tantalum monnitride				
0	---	1	N (1) Ta (1)	194.957
TaO₂				

2	---	1	O (1) Ta (1)	196.949
TaO2+1				
1	---	1	O (2) Ta (1)	212.949
TaO2 (s)				
0	---	1	O (2) Ta (1)	212.949
TaO2Cl (s) Tantalum chloride dioxide				
0	---	1	Cl (1) O (2) Ta (1)	248.402
TaOCl2 (s) Tantalum Oxychloride				
0	---	1	Cl (2) O (1) Ta (1)	267.855
TaOCl3 (s) Tantalum trichloride oxide				
0	---	1	Cl (3) O (1) Ta (1)	303.308
Tartaric-2 Tartrate ion				
-2	---	393	C (4) H (4) O (6)	148.072